

Earliest Stable Arrival Guidance for Explicit Stable-Completion Timing in Fixed-Horizon Diffusion Motion Planning

Abstract—Fixed-horizon diffusion motion planning can generate complete point-to-point trajectories, yet terminal goal constraints do not explicitly regulate when the task-completion state is established within the planning horizon. To address this issue, this paper proposes Earliest Stable Arrival Guidance (ESAG), which defines task completion through the persistent satisfaction of goal-position and low-velocity conditions over the remaining trajectory suffix and treats the stable-completion time as an explicit temporal optimization quantity within the fixed horizon. We further establish a suffix-occupancy representation of the earliest stable-arrival time, prove its exact equivalence to the normalized stable-arrival time, and derive a differentiable ESAG objective through continuous smoothing. This objective is incorporated into the reverse-diffusion process as inference-time guidance without retraining the diffusion prior or modifying the planning horizon. Experiments on 60 held-out scenarios covering regular, long-detour, and terminal-stabilization-challenging tasks show that ESAG increases the stable-completion rate from 9.00% to 91.33% while largely preserving the navigation success rate. It also advances both goal-entry and stable-arrival times, shortens the path length, and maintains a 100% collision-free rate together with high motion feasibility. Real-robot experiments further show that ESAG increases the stable-completion rate from 13.9% to 95.5% and exhibits a consistent advancement in stable-arrival timing.

Note to Practitioners— In many automation tasks, reaching the target position does not necessarily mean that the robot is ready to begin the next operation. For example, in indoor transportation, robotic docking, and laboratory logistics, the robot often needs to reduce its motion and remain stable before loading, docking, manipulation, or experimental operations can be reliably initiated. ESAG is designed to address this practical requirement by enabling the robot to establish and maintain a stable completion state earlier along the planned trajectory, thereby reducing the waiting time between navigation and subsequent operations and improving task-transition efficiency. The method can be integrated into an existing diffusion-based motion planner without retraining the original model or changing the prescribed planning duration, while still accounting for obstacle avoidance and motion constraints. The current method mainly targets point-to-point mobile-robot tasks with explicitly defined goal-position and stopping requirements. Dynamic environments and tasks with more complex completion conditions require further investigation. Future extensions may include mobile manipulation, multi-robot coordination, and other automation scenarios that require seamless transitions from navigation to subsequent operations.

Index Terms—Diffusion motion planning, earliest stable arrival, inference-time guidance, trajectory generation, fixed-horizon optimization, robotic automation.

I. INTRODUCTION

Motion planning is a key component connecting perception, decision-making, and execution in robotic automation systems, with broad applications in industrial manipulation [1], autonomous mobile robots [2], intelligent warehousing and logistics [3], [4], as well as autonomous scientific laboratories

and mobile laboratory robots [5], [6]. A motion planner is expected to connect a given initial state to a desired goal state while satisfying requirements on collision avoidance, motion feasibility, and execution efficiency. In recent years, diffusion models have provided a new paradigm for generative motion planning owing to their capability of modeling multimodal trajectory distributions. By learning structured motion priors from offline data and incorporating task objectives and planning constraints during inference, diffusion-based planners can adapt to new planning conditions while preserving multiple plausible motion modes [7]–[12].

The temporal semantics of diffusion-based planning depend on the underlying trajectory representation. Action-diffusion policies, visual-navigation methods, and autonomous-driving planners typically generate finite-length local actions or future trajectories and continuously replan in closed loop, such that a single generation window does not correspond to the duration of the complete task [13]–[15]. This work focuses on fixed-horizon point-to-point diffusion planning, where a complete motion is represented over a prescribed physical-time support with a fixed trajectory length and temporal discretization. Under this representation, terminal goal constraints can ensure eventual goal satisfaction, but do not explicitly specify when a stable task-completion state should be established within the planning horizon. Moreover, when the training trajectories share a common temporal support, the learned diffusion prior implicitly captures the statistical structure of motion progress along the physical-time indices, causing the formation time of stable completion to be influenced largely by the learned data distribution rather than by an explicit temporal objective. This issue is particularly important in autonomous logistics, laboratory automation, and robotic docking, where first entering the goal region does not necessarily imply task completion. The robot must also reach the target with sufficiently low residual motion and maintain this state throughout the subsequent trajectory; the final approach velocity itself can directly affect the reliability of subsequent connection and operation [16]. Therefore, for point-to-point motions with explicit terminal-state requirements, task completion naturally has a trajectory-suffix semantics: from a certain time onward, the goal-position and low-velocity conditions should remain satisfied over the remaining trajectory. This leads to the central question considered in this work: within a given fixed planning horizon, what is the earliest time from which the robot has stably completed the task and can maintain this completion state throughout the remaining trajectory.

Existing studies have introduced temporal regulation from different perspectives. Classical minimum-time planning and time-optimal path parameterization reduce execution time by adjusting the temporal scaling or terminal time of a trajec-

tory [17], [18]. VH-Diffuser treats the trajectory horizon as an instance-dependent variable and predicts the required trajectory length from the initial and goal states [19]. Multi-Robot Rolling Diffusion extends long-range multi-robot navigation through rolling-horizon generation and urgency-based temporal conditioning, allowing the progress of generated paths to be regulated over time [20]. Temporal Diffusion Planner reduces repeated replanning overhead in closed-loop diffusion planning by distributing denoising computation across successive decision steps [21]. In addition, zero-shot trajectory planning based on signal temporal logic explicitly handles externally specified progress and temporal constraints and leverages pretrained diffusion models to generate trajectories satisfying temporal specifications [22]. These approaches respectively address trajectory time scaling, horizon length, rolling temporal coverage, replanning computation scheduling, or externally imposed temporal constraints. In contrast, this work considers a complementary temporal quantity: with the trajectory length H , temporal discretization Δt , and maximum planning horizon $T_{\max}$ all kept fixed, we explicitly optimize the onset of stable task completion within the prescribed temporal support, referred to as the *completion onset*.

To address this problem, we propose ESAG for fixed-horizon diffusion motion planning. ESAG defines task completion as the persistent satisfaction of goal-position and low-velocity conditions over a trajectory suffix, and reformulates the earliest stable-arrival time as the suffix-wise occupancy of incompleteness. Based on this representation, a differentiable temporal objective is constructed and incorporated into the reverse-diffusion process, enabling explicit regulation of stable-completion timing without retraining the diffusion prior or modifying the planning horizon, while preserving collision safety and motion feasibility. The main contributions of this work are summarized as follows:

- We formulate the earliest stable-arrival problem for fixed-horizon diffusion motion planning, where task completion is characterized by the persistent satisfaction of goal-position and low-velocity conditions over the remaining trajectory suffix, and the stable-completion time is treated as an explicit temporal optimization quantity within the fixed planning horizon.
- We establish a suffix-occupancy representation of the earliest stable-arrival time, prove the exact equivalence between suffix-wise incompleteness occupancy and the normalized earliest stable-arrival time, and further construct a continuously differentiable relaxation that preserves this temporal semantics, enabling the originally discrete stable-completion time to directly participate in gradient-based optimization.
- We develop ESAG as an inference-time guidance mechanism that regulates stable-completion timing without retraining the diffusion prior or modifying the planning horizon. Simulation and real-robot experiments demonstrate that ESAG substantially improves the stable-completion rate and advances the stable-arrival time while preserving planning safety and motion feasibility.

II. RELATED WORK

A. Diffusion-Based Motion Planning and Inference-Time Trajectory Guidance

Diffusion models have become an important tool for representing multimodal trajectory distributions in generative motion planning. Their main advantage lies in the ability to learn structured motion priors from offline data and adapt generated trajectories to current task conditions during sampling. Diffuser models state-action trajectories as generative variables and enables goal-directed behavior generation through conditional inpainting and reward-based guidance [7]. In robotic motion planning, Motion Planning Diffusion (MPD) learns multimodal motion priors from precomputed feasible trajectories and modifies generated samples during reverse diffusion using gradients of goal, collision, and other planning costs, allowing the learned data prior and the current planning requirements to jointly influence the same sampling process [8], [9]. Unlike approaches that use generative models merely to provide trajectory initialization, such inference-time guidance directly acts on the denoising process and can therefore adapt to new planning objectives and constraints without retraining the trajectory prior.

Building on this paradigm, existing studies have extended the inference-time adaptation capability of diffusion planning through cost shaping, explicit constraints, and hybrid planning mechanisms. EDMP jointly guides diffusion sampling using multiple scene-dependent costs to balance collision avoidance and trajectory diversity [23]. Potential-Based Diffusion Motion Planning learns composable trajectory potential functions and converts different motion constraints into additive generative potentials [24]. SafeDiffuser incorporates safety specifications into the denoising process [25]. More recently, CoDiG introduces constraint-aware guidance through barrier functions during diffusion sampling [26], while Diffusion Predictive Control with Constraints handles out-of-distribution state and action constraints through model projection and constraint tightening [27]. JM²D formulates the interaction between a diffusion planner and a model-based optimization module as a joint sampling problem to improve compatibility between multimodal generation and safety/constraint modules [28]. For nonconvex or nondifferentiable objectives, Tree-Guided Diffusion Planner combines tree-structured exploration with conditional denoising to enhance test-time planning capability [29]. In addition, M²Diffuser, GPD, and multi-robot diffusion planning extend this framework toward high-dimensional mobile manipulation, trajectory parameterization, and multi-robot joint constraints, respectively [30]–[33].

Collectively, these studies demonstrate that diffusion priors can adapt to new task requirements at inference time through mechanisms such as cost guidance, projection, potential functions, and search. Their primary focus, however, is on specifying what conditions the generated trajectory should satisfy, while the formation time of a stable task-completion state along the physical-time axis remains insufficiently modeled and directly optimized.

B. Temporal Representation and Horizon Adaptation in Diffusion Planning

A fixed sequence length can have different temporal semantics across diffusion-based robotic methods. Closed-loop policies such as Diffusion Policy and NoMaD generate finite-length local actions or trajectories and continuously execute and replan as observations are updated. Therefore, a single generation window represents only a local prediction range rather than the duration of the complete task [13], [14]. Similarly, DiffusionDrive and Diffusion Planner generate vehicle trajectories over a finite future window and use them within continuous closed-loop planning [15], [34]. SICNav-Diffusion also incorporates diffusion-based trajectory prediction into a receding-horizon navigation framework, where repeated replanning enables continuous navigation in dynamic human-populated environments [35]. Therefore, a local closed-loop prediction window should be clearly distinguished from a fixed planning horizon representing a complete point-to-point motion.

When the duration of the complete task varies, existing diffusion-based methods mainly adapt by changing the generation horizon or extending temporal coverage. VH-Diffuser treats the trajectory horizon as a task-dependent variable and predicts an instance-specific trajectory length from the initial and goal states [19]. Multi-Robot Rolling Diffusion (MRRD) continuously generates local trajectories over a rolling horizon to cover long-range navigation tasks with different path lengths [20]. Another line of hierarchical methods decomposes long-horizon tasks into multiple local generation processes through key poses, subgoals, or skills. For example, ChainedDiffuser, HDP, and CHD address long-horizon task decomposition from the perspectives of key-pose prediction, hierarchical policies, and coupled high- and low-level diffusion, respectively [36]–[38].

Overall, existing time-related diffusion methods mainly focus on planning-horizon adaptation, long-horizon temporal coverage, and externally specified temporal constraints, while the formation time of a stable task-completion state within a fixed complete trajectory remains insufficiently modeled and explicitly optimized.

C. Time-Efficient Motion Planning and Stable Task Completion

Execution time has long been an important optimization objective in robotic motion planning. For a given geometric path, time-optimal path parameterization reduces execution time by reconfiguring the velocity evolution along the path. Verscheure *et al.* formulate time-optimal path tracking as a convex optimization problem [39]. Pham develops a general implementation of time-optimal path parameterization (TOPP) and subsequently proposes TOPP-RA based on reachability analysis [17], [18]. Subsequent studies further incorporate jerk constraints into time-optimal parameterization to obtain velocity profiles that better satisfy practical execution requirements [40]. These approaches mainly improve execution efficiency by adjusting the temporal scaling or terminal time of a trajectory, and therefore optimize a different temporal

quantity from the task-completion time considered within a fixed temporal support.

For autonomous logistics, scientific laboratory automation, and robotic docking tasks, the navigation goal is often not an isolated geometric position, but rather an interface to subsequent loading, equipment interaction, manipulation, or docking operations. Studies on mobile robotic chemists and self-driving laboratories show that mobile robots need to autonomously move among multiple experimental workstations and form continuous workflows with subsequent experimental operations [5], [6]. In more direct goal-docking tasks, Yadegar *et al.* further treat the final approach velocity as an important control variable during docking and regulate the desired closing velocity while performing obstacle avoidance [16]. These applications indicate that entering the goal region is not equivalent to reaching a terminal state from which the subsequent task can be initiated, and residual motion near the goal therefore has explicit task-level significance.

Accordingly, for point-to-point planning tasks with explicit terminal-state requirements, using only the goal distance or the first entry into the goal region as the completion criterion is insufficient. The robot may traverse the goal region at high speed or leave it after only transiently satisfying the goal condition. A task-completion state that is consistent with subsequent automation operations should satisfy both positional and low-velocity requirements and remain valid throughout the subsequent trajectory. Task completion therefore naturally exhibits trajectory-suffix semantics, requiring the identification of the physical-time index from which all remaining trajectory states continuously satisfy the prescribed completion conditions. Overall, existing studies improve planning performance and temporal efficiency through trajectory-constraint guidance, horizon adaptation, rolling planning, time parameterization, explicit temporal specifications, and execution-time scaling. However, for fixed-horizon diffusion planning of complete point-to-point motions, a unified mechanism is still lacking to jointly characterize whether stable completion has been established and when it begins to hold persistently. This constitutes the central problem addressed by ESAG.

III. PROBLEM FORMULATION

A. Energy-Guided Planning under a Diffusion Trajectory Prior

Diffusion-based motion planning can be formulated as a probabilistic inference process jointly governed by a learned trajectory prior and task-related energies. The diffusion model captures multimodal motion distributions from offline trajectories, while online planning objectives condition the reverse generation process through energy gradients. It is important to distinguish the physical time dimension of a robot trajectory from the independent diffusion dimension introduced by the generative process. Accordingly, we use $t \in \{0, \dots, H-1\}$ to denote the physical-time index and $n \in \{0, \dots, N\}$ to denote the diffusion level. The clean trajectory is denoted by $\tau^{(0)}$, and its random variable at the n th diffusion level is denoted by $\tau^{(n)}$. All task-completion and arrival times considered hereafter are defined with respect to the physical-time index

t . During training, the diffusion model progressively perturbs demonstration trajectories with Gaussian noise and learns to recover structured motions from their corrupted counterparts. The marginal forward diffusion process is given by

$$q(\boldsymbol{\tau}^{[n]} | \boldsymbol{\tau}^{[0]}) = \mathcal{N}(\sqrt{\bar{\alpha}_n} \boldsymbol{\tau}^{[0]}, (1 - \bar{\alpha}_n) \mathbf{I}), \quad (1)$$

where $\alpha_n = 1 - \beta_n$, $\bar{\alpha}_n = \prod_{i=1}^n \alpha_i$, and $\{\beta_n\}_{n=1}^N$ denotes a prescribed variance schedule. Equivalently, a trajectory at diffusion level n can be sampled as

$$\boldsymbol{\tau}^{[n]} = \sqrt{\bar{\alpha}_n} \boldsymbol{\tau}^{[0]} + \sqrt{1 - \bar{\alpha}_n} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

A denoising network ϵ_θ is trained to estimate the noise injected by the forward diffusion process:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\boldsymbol{\tau}^{[0]}, n, \boldsymbol{\epsilon}} \left[\left\| \boldsymbol{\epsilon} - \epsilon_\theta(\boldsymbol{\tau}^{[n]}, n) \right\|_2^2 \right]. \quad (3)$$

After training, the model does not represent a unique solution to a specific planning instance. Instead, it captures a probability distribution over structured motions induced by the demonstration data. The learned reverse transition can be written as

$$p_\theta(\boldsymbol{\tau}^{[n-1]} | \boldsymbol{\tau}^{[n]}) = \mathcal{N}(\boldsymbol{\mu}_{\theta, n}, \boldsymbol{\Sigma}_n). \quad (4)$$

This observation is important for the planning formulation considered here: the diffusion model provides a *motion prior*, rather than a complete solution that already satisfies all online constraints associated with the current planning instance. The reverse sampling process must therefore be further shaped according to the task-specific requirements encountered at inference time.

Given $\boldsymbol{\tau}^{[n]}$, the denoising network provides an estimate of the corresponding clean trajectory,

$$\hat{\boldsymbol{\tau}}_n^{[0]} = \frac{\boldsymbol{\tau}^{[n]} - \sqrt{1 - \bar{\alpha}_n} \epsilon_\theta(\boldsymbol{\tau}^{[n]}, n)}{\sqrt{\bar{\alpha}_n}}. \quad (5)$$

Unlike the noisy variable in diffusion space, $\hat{\boldsymbol{\tau}}_n^{[0]}$ recovers the physical semantics of the robot state sequence. Collision costs, trajectory smoothness, motion constraints, and the stable task-completion objective introduced later can therefore be evaluated directly on this clean estimate. In this sense, (5) establishes the interface between the iterative denoising process in diffusion space and planning objectives defined over physically meaningful robot trajectories.

Based on this connection, we formulate online motion planning as an energy-based reweighting of the learned trajectory prior. Let $\mathcal{P}$ denote the conditions associated with the current planning instance, including the initial state, goal state, environment information, and motion constraints. Let $J(\boldsymbol{\tau}; \mathcal{P})$ denote a differentiable planning energy jointly determined by the trajectory and these conditions. The desired task-conditioned trajectory distribution can then be expressed as

$$p(\boldsymbol{\tau} | \mathcal{P}) \propto p_\theta(\boldsymbol{\tau}) \exp[-J(\boldsymbol{\tau}; \mathcal{P})]. \quad (6)$$

Equation (6) provides the probabilistic basis for the guidance mechanism used throughout this work. Rather than replacing the diffusion model with an independent optimizer, the planning energy J reweights the learned motion prior

according to the current task. Trajectories that are both plausible under the learned prior and consistent with the online planning requirements receive higher posterior probability, whereas trajectories that conflict with the task conditions are suppressed.

Taking the gradient of the log-density in (6) yields

$$\nabla_{\boldsymbol{\tau}} \log p(\boldsymbol{\tau} | \mathcal{P}) = \nabla_{\boldsymbol{\tau}} \log p_\theta(\boldsymbol{\tau}) - \nabla_{\boldsymbol{\tau}} J(\boldsymbol{\tau}; \mathcal{P}). \quad (7)$$

As can be seen from (7), the score of the conditional trajectory distribution is jointly determined by the learned prior and the task-related energy. The former preserves the motion structure encoded in the demonstration data, while the latter provides task-dependent corrections according to the current planning conditions. This decomposition allows different planning attributes to be incorporated into trajectory generation in a unified manner through differentiable energy terms, without modifying the learned trajectory prior itself.

The same probabilistic decomposition also provides a unified basis for modeling task-completion time under a fixed temporal representation. In addition to goal reachability, collision avoidance, and motion feasibility, the temporal completion characteristics of a trajectory can likewise be reflected in the conditional distribution through an appropriate energy formulation. Based on this perspective, the following section defines the temporal semantics of stable task completion and develops a differentiable representation suitable for continuous generative optimization.

B. Stable Completion Time under a Fixed-Horizon Representation

Under the probabilistic inference framework described above, we further consider task-completion time within a fixed planning horizon. This work focuses on fixed-horizon diffusion planning for complete point-to-point motions, where the clean trajectory is represented as

$$\boldsymbol{\tau}^{(0)} = (\mathbf{s}_0, \mathbf{s}_1, \dots, \mathbf{s}_{H-1}), \quad \mathbf{s}_t = (\mathbf{q}_t, \dot{\mathbf{q}}_t), \quad (8)$$

where H denotes the number of trajectory support points and Δt is the fixed temporal discretization interval. The corresponding planning horizon is therefore

$$T_{\max} = (H - 1)\Delta t. \quad (9)$$

Throughout this work, H , Δt , and $T_{\max}$ remain fixed. Consequently, all candidate motions are defined over the same physical temporal support, rather than adapting the task duration by changing the trajectory length or temporal scale.

For a given point-to-point planning task, the trajectory satisfies

$$\mathbf{s}_0 = \mathbf{s}_{\text{start}}, \quad \mathbf{q}_{H-1} \in \mathcal{G}_q, \quad (10)$$

where $\mathcal{G}_q$ denotes the goal-position set. The terminal condition ensures that the trajectory reaches the prescribed goal region at the end of the planning horizon, but does not guarantee that a stable task-completion state has been established. Therefore, trajectories sharing the same terminal state may still exhibit different actual completion times.

Meanwhile, the first entry into the goal region does not, by itself, sufficiently characterize task completion. The robot may

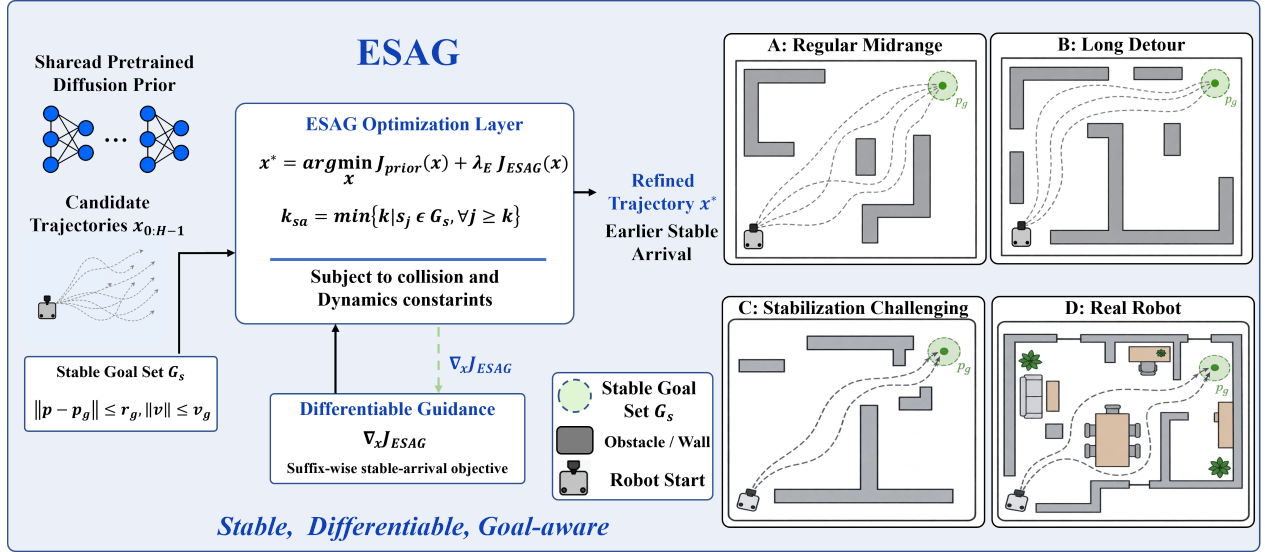


Fig. 1. Overview of the proposed ESAG framework. A differentiable earliest-stable-arrival objective is incorporated into reverse diffusion to guide trajectory refinement under collision and motion-feasibility constraints. Representative planning scenarios are shown on the right.

traverse the goal region at a relatively high velocity or leave the region after only transiently satisfying the goal condition. We therefore characterize stable completion as a trajectory-suffix property and define the earliest stable arrival index as

$$k_{sa}(\tau) = \min \{k \mid s_t \in \mathcal{G}_s, \forall t \in \{k, \dots, H-1\}\}. \quad (11)$$

The corresponding earliest stable arrival time is

$$T_{sa}(\tau) = k_{sa}(\tau) \Delta t. \quad (12)$$

Accordingly, the temporal optimization problem considered in this work can be formulated as

$$\min_{\tau \in \mathcal{F}(\mathcal{P})} T_{sa}(\tau), \quad (13)$$

where $\mathcal{F}(\mathcal{P})$ denotes the set of trajectories satisfying the safety and motion-feasibility requirements associated with the current planning instance. Since the temporal support remains fixed, the above objective essentially characterizes the earliest establishment of a stable task-completion state within the prescribed horizon, rather than optimizing the terminal time of the trajectory itself.

However, $k_{sa}(\tau)$ simultaneously involves state-set membership, persistent satisfaction over a trajectory suffix, and a discrete minimum index. It is therefore inherently nonsmooth and cannot be directly used as a gradient-based guidance objective. To address this issue, we next establish an equivalent suffix-occupancy representation and use it to construct a continuously differentiable earliest-stable-arrival objective.

IV. EARLIEST STABLE ARRIVAL GUIDANCE

A. ESAG Framework and Algorithmic Overview

ESAG is developed for fixed-horizon diffusion motion planning and introduces earliest-stable-arrival guidance to explicitly regulate the time at which the task-completion state is established within the planning horizon. As illustrated in Fig. 1, the method generates candidate trajectories from a pretrained

diffusion trajectory prior shared with the baseline planner and incorporates a differentiable ESAG temporal objective into the reverse-diffusion process. This objective is jointly considered with baseline planning constraints, including collision avoidance and motion feasibility, so that the generated trajectories can establish stable task completion earlier while preserving safety and dynamic feasibility. ESAG consists of three closely related components: the stable goal set, the suffix-occupancy representation, and the differentiable earliest-stable-arrival objective. The stable goal set defines the task-completion state; the suffix-occupancy representation characterizes the persistent satisfaction of this state over the remaining trajectory suffix and thereby provides a trajectory-level representation of the earliest stable-arrival time. A differentiable ESAG objective is then constructed through continuous smooth approximations and its gradient is incorporated into the reverse-diffusion updates. The detailed definitions and theoretical relations of these components are presented in the following subsections, while the overall planning procedure is summarized in Algorithm 1.

B. Stable Goal-State Set

The preceding section introduced $\mathcal{G}_s$ as an abstract task-completion state set. We now specify its construction. For point-to-point motion planning, task completion cannot in general be determined solely by whether the robot enters the goal region, since the robot may traverse that region at a relatively high velocity. In such a case, the positional condition is satisfied only instantaneously, while the motion cannot yet be regarded as validly completed.

Let $\mathcal{Q}$ denote the robot configuration space, let $\mathbf{q}_g$ denote the goal configuration, and let $d_g(\mathbf{q})$ denote the signed distance from configuration $\mathbf{q}$ to the goal region. The goal-position set is defined as

$$\mathcal{G}_q = \{\mathbf{q} \in \mathcal{Q} \mid d_g(\mathbf{q}) \leq 0\}. \quad (14)$$

Algorithm 1 ESAG-Guided Diffusion Motion Planning

Require: Trajectory dataset $\mathcal{D}$, planning condition $\mathcal{P}$, stable goal set $\mathcal{G}_s$, horizon H , reverse steps N , guidance weight λ_E

Ensure: Planned trajectory τ^*

Offline: Shared diffusion prior learning

- 1: Train the diffusion prior p_θ on $\mathcal{D}$ using the standard diffusion objective
- 2: Freeze the pretrained parameters θ

Online: ESAG-guided planning

- 3: Sample $\tau^{[N]} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and impose the prescribed boundary conditions
- 4: **for** $n = N, \dots, 1$ **do**
- 5: $\hat{\tau}_n^{[0]} \leftarrow p_\theta(\tau^{[n]}, n, \mathcal{P})$
- 6: $J_{\text{base}} \leftarrow J_{\text{base}}(\hat{\tau}_n^{[0]})$
- 7: $\hat{b}_t \leftarrow \hat{b}_t(\hat{\tau}_n^{[0]}; \mathcal{G}_s), \quad t = 0, \dots, H-1$
- 8: $\hat{u}_t \leftarrow \text{SufAgg}(\{\hat{b}_j\}_{j=t}^{H-1}), \quad t = 0, \dots, H-2$
- 9: $J_{\text{ESAG}} \leftarrow \frac{1}{H-1} \sum_{t=0}^{H-2} \hat{u}_t$
- 10: $\mathbf{g}_n \leftarrow \nabla_{\tau^{[n]}} (J_{\text{base}} + \lambda_E J_{\text{ESAG}})$
- 11: $\tilde{\tau}^{[n-1]} \leftarrow \text{GuideStep}(\tau^{[n]}, \hat{\tau}_n^{[0]}, -\mathbf{g}_n)$
- 12: $\tau^{[n-1]} \leftarrow \text{FeasibleUpdate}_{\mathcal{F}}(\tilde{\tau}^{[n-1]})$
- 13: Reimpose the prescribed boundary conditions
- 14: **end for**
- 15: $\tau^* \leftarrow \text{Select}_{\mathcal{F}}(\tau^{[0]})$
- 16: **return** τ^*

For a spherical goal region centered at $\mathbf{q}_g$ with tolerance radius r_g ,

$$d_g(\mathbf{q}) = \|\mathbf{q} - \mathbf{q}_g\|_2 - r_g. \quad (15)$$

Let $v_g > 0$ denote the maximum velocity magnitude admissible for task completion. For a state $\mathbf{s} = (\mathbf{q}, \dot{\mathbf{q}})$, the stable goal-state set is defined as

$$\mathcal{G}_s = \{(\mathbf{q}, \dot{\mathbf{q}}) \mid d_g(\mathbf{q}) \leq 0, \|\dot{\mathbf{q}}\|_2 \leq v_g\}. \quad (16)$$

Accordingly,

$$\mathbf{s}_t \in \mathcal{G}_s \quad (17)$$

indicates that the trajectory state at physical index t simultaneously satisfies the prescribed positional tolerance and completion-velocity requirement. The purpose of $\mathcal{G}_s$ is not to introduce an additional stopping task independent of the original planning objective, but to specify which robot states can legitimately be regarded as valid task-completion states. Consequently, high-speed passages through the goal region are excluded from stable completion.

The terminal goal condition requires the final configuration to lie in the goal-position set,

$$\mathbf{q}_{H-1} \in \mathcal{G}_q. \quad (18)$$

This positional terminal condition does not necessarily imply $\mathbf{s}_{H-1} \in \mathcal{G}_s$, since the residual velocity may still exceed the stable-completion threshold. ESAG therefore explicitly regulates when a persistently stable completion state is established within the fixed temporal support.

C. Suffix-Occupancy Representation and Arrival-Time Equivalence

The earliest stable arrival index $k_{\text{sa}}(\tau)$ and the corresponding physical time $T_{\text{sa}}(\tau)$ have already been defined in the preceding section. Rather than directly approximating this discrete minimum index, ESAG considers an equivalent question: at each physical trajectory index, has the task already entered a persistently completed state?

We first define the instantaneous violation indicator as

$$b_t(\tau) = \mathbb{I}[\mathbf{s}_t \notin \mathcal{G}_s] = \begin{cases} 0, & \mathbf{s}_t \in \mathcal{G}_s, \\ 1, & \mathbf{s}_t \notin \mathcal{G}_s. \end{cases} \quad (19)$$

The suffix violation beginning at physical index t is then defined as

$$u_t(\tau) = \max_{j \in \{t, \dots, H-1\}} b_j(\tau). \quad (20)$$

By construction, $u_t(\tau) = 0$ if and only if every state in the trajectory suffix beginning at t belongs to $\mathcal{G}_s$. Hence, u_t can be interpreted as a binary indicator of whether stable task completion has not yet been established at physical index t .

This observation motivates the ideal ESAG cost

$$J_{\text{ESAG}}^{\text{hard}}(\tau) = \frac{1}{H-1} \sum_{t=0}^{H-2} u_t(\tau). \quad (21)$$

Rather than directly penalizing a discrete arrival index, (21) measures the normalized temporal occupancy during which stable task completion has not yet been established. An earlier onset of the stable-completion suffix therefore produces fewer nonzero suffix violations and a smaller cost.

Proposition 1 (Suffix-Occupancy Equivalence). *Suppose that the terminal state satisfies $\mathbf{s}_{H-1} \in \mathcal{G}_s$. Then*

$$J_{\text{ESAG}}^{\text{hard}}(\tau) = \frac{k_{\text{sa}}(\tau)}{H-1} = \frac{T_{\text{sa}}(\tau)}{T_{\text{max}}}. \quad (22)$$

Proof. By definition of $k_{\text{sa}}(\tau)$, every suffix beginning before $k_{\text{sa}}(\tau)$ contains at least one state outside $\mathcal{G}_s$, whereas every suffix beginning at or after $k_{\text{sa}}(\tau)$ is entirely contained in $\mathcal{G}_s$. Therefore,

$$u_t(\tau) = \begin{cases} 1, & 0 \leq t < k_{\text{sa}}(\tau), \\ 0, & k_{\text{sa}}(\tau) \leq t \leq H-1. \end{cases} \quad (23)$$

It follows that

$$\sum_{t=0}^{H-2} u_t(\tau) = k_{\text{sa}}(\tau). \quad (24)$$

Together with $T_{\text{sa}} = k_{\text{sa}}\Delta t$ and $T_{\text{max}} = (H-1)\Delta t$, substitution into (21) yields (22).

Proposition 1 establishes that the earliest stable-arrival time within a fixed planning horizon can be represented exactly by the temporal occupancy of suffix-wise incompleteness. As illustrated in Fig. 2, the instantaneous violation indicator b_t may temporarily become zero before stable completion is actually established, whereas the suffix violation u_t remains nonzero as long as any future violation exists. Consequently, u_t switches to zero only at k_{sa} and remains zero thereafter, preserving the persistence semantics of stable task completion.

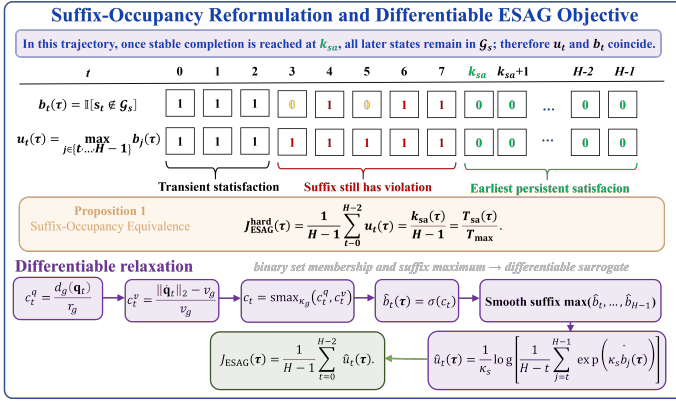


Fig. 2. Suffix-occupancy reformulation and differentiable ESAG objective. The instantaneous violation indicator b_t describes stable-goal-set satisfaction at each trajectory step, whereas the suffix violation u_t encodes whether any future violation remains. The resulting suffix occupancy is exactly equivalent to the normalized earliest stable-arrival time. Its smooth relaxation yields the differentiable ESAG objective used for gradient-based guidance.

D. Differentiable ESAG Objective

Although $J_{\text{ESAG}}^{\text{hard}}$ has an exact temporal interpretation, the state-membership test and suffix maximum are both nonsmooth operations. To obtain a temporal energy suitable for continuous trajectory guidance, we construct smooth approximations of the instantaneous and suffix-wise violation measures.

1) *Smooth Stable-Goal Violation:* We first define normalized violations of the positional and completion-velocity conditions as

$$c_t^q = \frac{d_g(\mathbf{q}_t)}{r_g}, \quad (25)$$

and

$$c_t^v = \frac{\|\dot{\mathbf{q}}_t\|_2 - v_g}{v_g}. \quad (26)$$

The positional condition is satisfied when $c_t^q \leq 0$, whereas the completion-velocity condition is satisfied when $c_t^v \leq 0$. Since membership in $\mathcal{G}_s$ requires both conditions to hold simultaneously, we aggregate the two violations through the smooth maximum

$$c_t = \text{smax}_{\kappa_g}(c_t^q, c_t^v), \quad (27)$$

where

$$\text{smax}_{\kappa_g}(a, b) = \frac{1}{\kappa_g} \log \left(\frac{\exp(\kappa_g a) + \exp(\kappa_g b)}{2} \right), \quad \kappa_g > 0. \quad (28)$$

This approximation satisfies

$$\max(a, b) - \frac{\log 2}{\kappa_g} \leq \text{smax}_{\kappa_g}(a, b) \leq \max(a, b), \quad (29)$$

and therefore

$$\text{smax}_{\kappa_g}(a, b) \rightarrow \max(a, b) \quad \text{as } \kappa_g \rightarrow \infty. \quad (30)$$

Thus, c_t provides a smooth representation of the overall stable-goal violation. We then construct the soft state-violation measure

$$\hat{b}_t(\tau) = \sigma(\kappa_b c_t) = \frac{1}{1 + \exp(-\kappa_b c_t)}, \quad \kappa_b > 0. \quad (31)$$

For states away from the boundary of $\mathcal{G}_s$, the corresponding hard-limit behavior is

$$\hat{b}_t(\tau) \rightarrow \begin{cases} 0, & s_t \in \mathcal{G}_s, \\ 1, & s_t \notin \mathcal{G}_s. \end{cases} \quad (32)$$

The parameters κ_g and κ_b independently control the sharpness of the condition aggregation and the soft set-membership transition, respectively.

2) *Smooth Suffix Violation:* Given the continuous instantaneous violation measure, we next smooth the suffix maximum in (20). For each physical trajectory index t , define

$$\hat{u}_t(\tau) = \frac{1}{\kappa_s} \log \left[\frac{1}{H-t} \sum_{j=t}^{H-1} \exp(\kappa_s \hat{b}_j(\tau)) \right], \quad \kappa_s > 0, \quad (33)$$

where κ_s controls the sharpness of the suffix aggregation. The normalization factor $1/(H-t)$ removes the suffix-length-dependent offset introduced by the standard LogSumExp operator.

Let

$$m_t = \max_{j \in \{t, \dots, H-1\}} \hat{b}_j(\tau). \quad (34)$$

Then

$$m_t - \frac{\log(H-t)}{\kappa_s} \leq \hat{u}_t(\tau) \leq m_t. \quad (35)$$

Hence,

$$\hat{u}_t(\tau) \rightarrow \max_{j \in \{t, \dots, H-1\}} \hat{b}_j(\tau) \quad \text{as } \kappa_s \rightarrow \infty. \quad (36)$$

The differentiable ESAG energy is finally defined as

$$J_{\text{ESAG}}(\tau) = \frac{1}{H-1} \sum_{t=0}^{H-2} \hat{u}_t(\tau). \quad (37)$$

Unlike an instantaneous goal-attraction objective, J_{ESAG} accumulates a *soft suffix-wise incompleteness measure* over the entire temporal support. Reducing this energy therefore regulates the onset of the stable task-completion suffix along the fixed temporal axis, rather than merely decreasing the goal distance at an individual trajectory state.

For trajectories whose states do not lie on the boundary of $\mathcal{G}_s$, as κ_g , κ_b , and κ_s jointly approach their corresponding hard limits,

$$J_{\text{ESAG}}(\tau) \rightarrow J_{\text{ESAG}}^{\text{hard}}(\tau) = \frac{T_{\text{sa}}(\tau)}{T_{\text{max}}}. \quad (38)$$

Thus, for finite smoothing parameters, J_{ESAG} constitutes a continuously differentiable surrogate of the normalized earliest stable arrival time, while its hard-limit form recovers the exact discrete temporal interpretation established in Proposition 1.

Within the unified energy-guided planning formulation, ESAG enters as a temporal-completion component of the overall planning energy:

$$J_{\text{plan}}(\tau; \mathcal{P}) = J_{\text{base}}(\tau; \mathcal{P}) + \lambda_E J_{\text{ESAG}}(\tau), \quad \lambda_E \geq 0, \quad (39)$$

where J_{base} collects the collision, safety, and motion-feasibility energies associated with the current planning instance, and λ_E determines the relative contribution of the

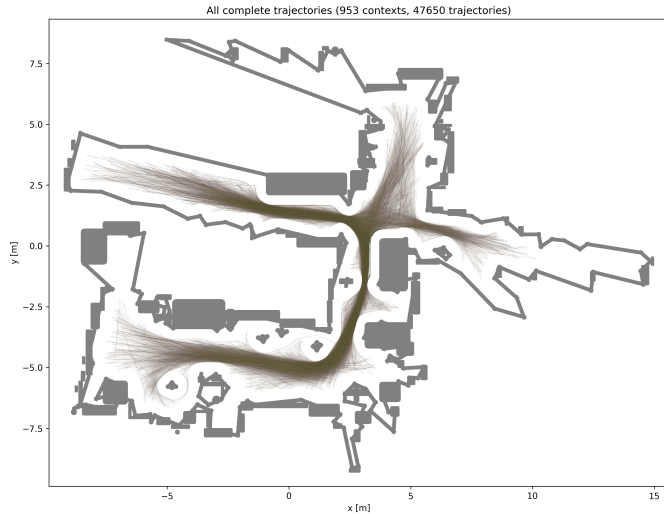


Fig. 3. Simulation environment and representative navigation tasks under different spatial constraints, including open areas, narrow passages, successive obstacle-avoidance regions, and constrained arrival regions with goals located near walls or laboratory equipment.

stable-completion-time objective. In this manner, a task-completion time originally defined through discrete suffix semantics is transformed into a differentiable energy component of the conditional trajectory distribution, completing the connection between fixed-horizon stable-completion modeling and energy-guided diffusion planning.

V. EXPERIMENTS AND RESULTS

A. Experimental Setup and Compared Methods

1) *Scene Stratification*: To systematically evaluate the planning performance of ESAG under different spatial constraints and task loads, the experiments are conducted in an indoor navigation environment with a representative scientific-laboratory layout, including open areas, narrow passages, consecutive obstacle-avoidance regions, and constrained goal-reaching regions adjacent to walls or laboratory equipment. Rather than varying task difficulty by simply increasing the number of randomly placed obstacles, this environment is designed to explicitly cover three complementary aspects, namely geometric detours, temporal load, and terminal stabilization, thereby distinguishing the ability to reach the goal from the ability to establish a stable completion state within a finite planning horizon.

The simulation employs an omnidirectional mobile robot model with dimensions consistent with those of the physical platform. The collision geometry of the mobile base is modeled using an approximately 0.35 m two-dimensional footprint together with an additional safety margin, while consistent translational, rotational, and acceleration constraints are imposed. The diffusion trajectory prior is trained on approximately 5×10^4 collision-free indoor navigation trajectories covering about 1,000 environment contexts, as illustrated in Fig. 3, using six NVIDIA RTX A6000 GPUs. MPD and ESAG share the same pretrained diffusion model and use identical trajectory lengths, temporal discretization,

and maximum planning horizons. Therefore, their performance difference directly reflects the effect of the earliest stable-arrival guidance introduced during inference.

The test set is constructed from held-out indoor navigation scenarios that are strictly separated from the training set, and none of the test scenarios is used for training the diffusion model. Based on pre-generated collision-free reference trajectories, the path detour ratio, temporal load, trajectory turning density, and terminal motion characteristics in the goal-approach phase are computed for each scenario. Prior to algorithm evaluation, 60 navigation scenarios are fixed and divided into three categories, with 20 scenarios in each category: Category A, *Regular Midrange*, represents regular medium-range navigation tasks; Category B, *Long Detour*, exhibits higher path-detour and temporal loads; and Category C, *Stabilization Challenging*, is primarily characterized by the difficulty of deceleration and stable terminal-state formation near the goal. The scenario selection and category assignment are determined once before evaluation and remain unchanged across all planning methods and repeated trials, ensuring that all methods are compared under identical task conditions.

As shown in Fig. 4, the three task categories exhibit clear differences in their representative scenario characteristics and dominant sources of difficulty. Figures 4(a)–(c) present representative examples of the three categories. Figure 4(d) further illustrates their distributions in the task-feature space, where Category A is concentrated in the low-task-load region, Category B is primarily separated along the global-detour difficulty dimension, and Category C is mainly distinguished along the stable-completion difficulty dimension. Figures 4(e) and 4(f) report the statistical distributions of the path detour ratio and the final-approach turn density, respectively. The mean path detour ratio of Category B is 2.208, which is substantially higher than those of Category A (1.048) and Category C (1.237). In contrast, the mean final-approach turn density of Category C reaches 0.676 rad/m, compared with 0.051 rad/m for Category A and 0.178 rad/m for Category B. Accordingly, the constructed test set covers three representative planning conditions, namely regular motion, global detour with increased temporal load, and terminal stable completion.

2) *Compared Methods*: All methods are evaluated using the same environment map, robot model, start–goal configurations, and motion constraints. Each task category contains 20 fixed scenarios, and each scenario is repeated five times, resulting in 300 trials per method and 1,200 trials in total for the four methods. RRT*+TOPP-RA, GPMP2-ATS, and MPD are selected as comparison methods. RRT*+TOPP-RA combines RRT* geometric path search [41] with TOPP-RA time-optimal path parameterization [18], representing a classical two-stage planning paradigm. GPMP2-ATS is built upon the continuous-time Gaussian-process trajectory representation and factor-graph optimization framework of GPMP2 [42], together with adaptive time search. MPD [8], [9] serves as the most directly related diffusion-planning baseline and shares with ESAG the same pretrained trajectory prior, planning horizon, temporal discretization, and candidate settings, enabling the effect of ESAG guidance to be directly assessed.

The evaluation metrics cover four aspects: task completion,

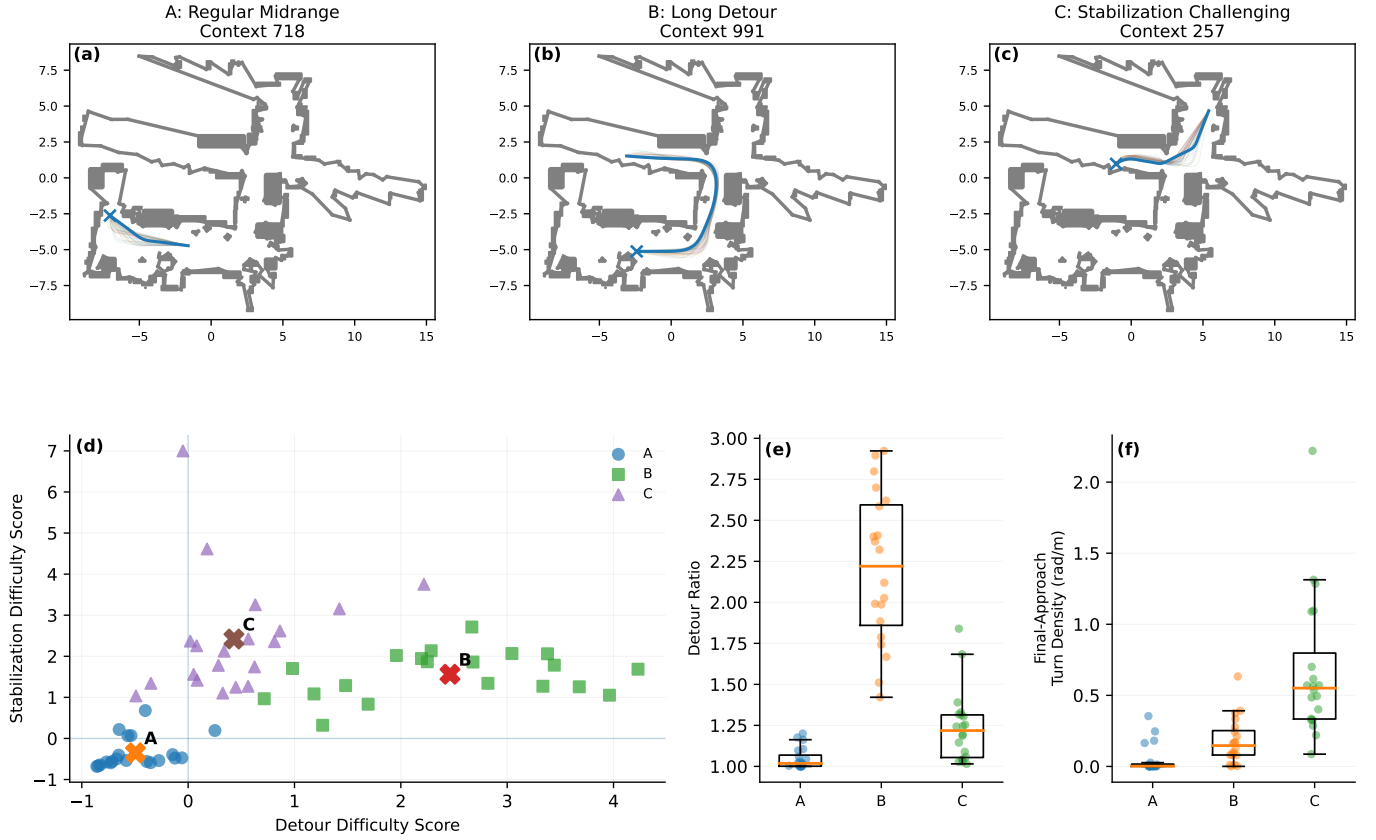


Fig. 4. Taxonomy and representative characteristics of the fixed evaluation scenarios. (a)–(c) Representative examples of Regular Midrange, Long Detour, and Stabilization Challenging tasks. (d) Distribution of the 60 scenarios in the task-difficulty space. (e) Path detour ratio and (f) final-approach turn density for the three categories.

TABLE I
OVERALL PLANNING PERFORMANCE ON 60 FIXED HELD-OUT SCENARIOS. EACH METHOD IS EVALUATED OVER 300 TRIALS.

Method	Nav. (%)	Stable (%)	CF (%)	Dyn. (%)	T_{enter} (s)	T_{sa} (s)	Path (m)	Planning (s)
RRT*+TOPP-RA	72.67	72.67	86.00	72.67	5.373	5.611	9.483	10.261
GPMP2-ATS	78.00	2.33	78.00	78.00	5.225	8.214	8.996	40.852
MPD	92.33	9.00	100.00	92.33	7.685	7.875	9.019	0.452
ESAG	91.67	91.33	100.00	94.67	6.052	6.438	7.947	0.685

motion efficiency, trajectory feasibility, and computational efficiency. Task-completion performance is evaluated using the navigation success rate (Nav.) and stable-completion rate (Stable). Temporal efficiency is measured by the first goal-entry time, T_{enter} , and the earliest stable-arrival time, T_{sa} , where T_{sa} requires the trajectory to satisfy the low-speed condition after entering the goal region and to continuously maintain the stable state throughout the remaining planning horizon. Trajectory feasibility is assessed using the collision-free rate (CF), dynamic-feasibility rate (Dyn.), maximum velocity, and maximum acceleration, while path length is additionally reported to characterize the geometric cost. Planning time is also reported to evaluate computational efficiency. The temporal metrics are computed only over successful trajectories satisfying the corresponding validity conditions.

B. Comparative Results

Table I summarizes the overall results over the 60 fixed scenarios. MPD and ESAG achieve navigation success rates of 92.33% and 91.67%, respectively, differing by only 0.66 percentage points, indicating that the earliest stable-arrival guidance largely preserves the geometric navigation capability of the original diffusion planner. In contrast, a substantial difference is observed in stable-completion performance: the stable-completion rate of MPD is only 9.00%, whereas ESAG achieves 91.33%, corresponding to an absolute improvement of 82.33 percentage points. If the ratio Stable/Nav. is used to measure the fraction of successfully navigated trajectories that eventually establish a stable terminal state, the corresponding values are approximately 9.75% for MPD and 99.63% for ESAG. This discrepancy indicates that the primary limitation of MPD is not the geometric reachability of the goal region,

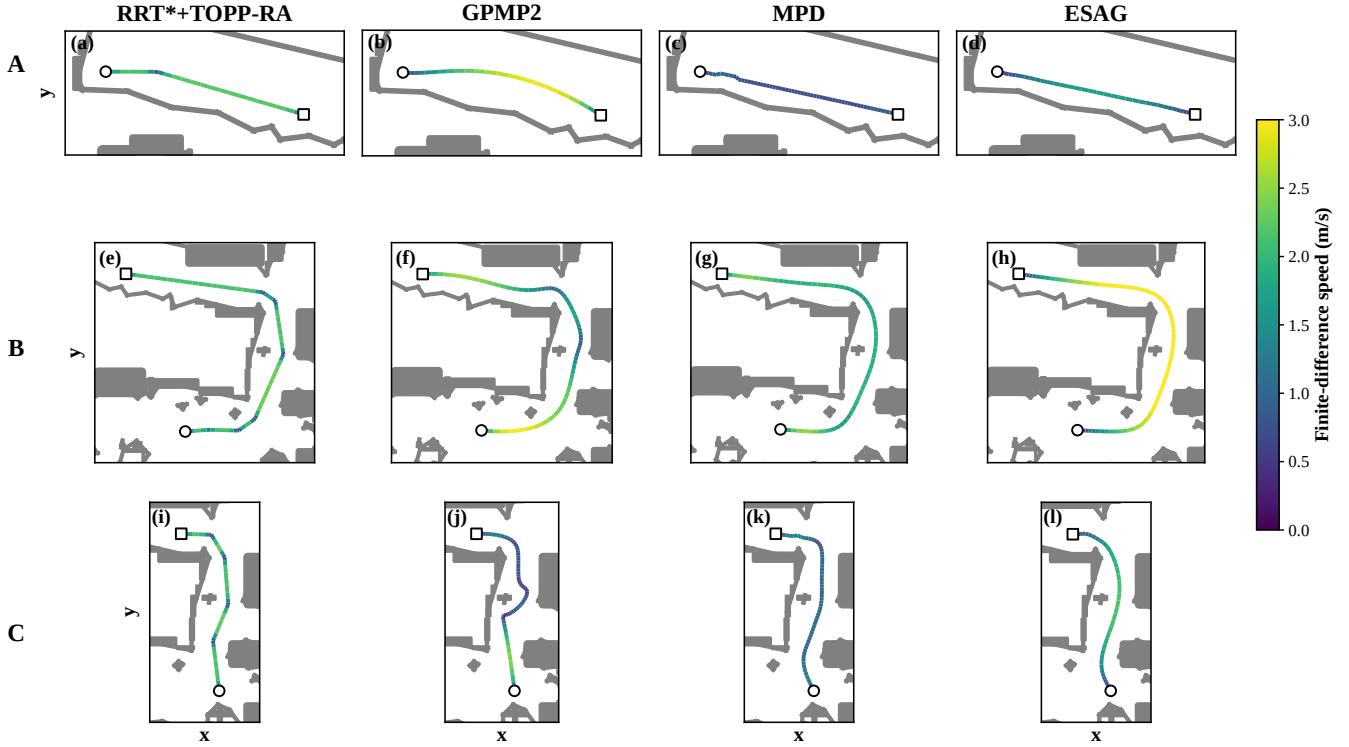


Fig. 5. Representative trajectories of RRT*+TOPP-RA, GPMP2-ATS, MPD, and ESAG on one fixed context from each scene category. Trajectory color denotes the finite-difference speed. The three rows correspond to Category A (context 305), Category B (context 795), and Category C (context 986), respectively. These examples correspond to representative individual trials rather than category-wise averages.

but rather its difficulty in converting goal-region entry into a persistently low-speed and maintainable task-completion state within the fixed planning horizon.

The temporal metrics further reveal the difference between the two diffusion planners in how motion is organized within the fixed planning horizon. For MPD, the mean T_{enter} and T_{sa} are 7.685 s and 7.875 s, respectively, indicating that stable completion is concentrated near the end of the planning horizon. ESAG reduces these values to 6.052 s and 6.438 s, advancing first goal entry by 1.633 s and stable completion by 1.437 s, respectively. Meanwhile, the average path length decreases from 9.019 m to 7.947 m, while both MPD and ESAG maintain a 100% collision-free rate. This shows that the improvement in stable-completion performance is not obtained through additional detouring or by sacrificing collision safety. The average planning time increases from 0.452 s for MPD to 0.685 s for ESAG, corresponding to an additional 0.233 s, while remaining substantially lower than those of the two conventional planning baselines.

The cross-paradigm comparison should be interpreted together with the different temporal mechanisms adopted by the planners. RRT*+TOPP-RA achieves both a navigation success rate and a stable-completion rate of 72.67%. Its two-stage time-parameterization procedure typically drives successfully generated trajectories to a stationary terminal state. GPMP2-ATS achieves a navigation success rate of 78.00%, but its stable-completion rate is only 2.33%, indicating that adaptive time search based on conventional geometric and dynamic feasibility does not necessarily guarantee a persistently low-

speed terminal state. Since RRT*+TOPP-RA and GPMP2-ATS retain free-time parameterization, whereas MPD and ESAG use a unified fixed planning horizon, the cross-paradigm temporal metrics mainly characterize their overall planning behavior. In contrast, the strictly matched configurations of MPD and ESAG enable a more direct assessment of the contribution of the ESAG guidance itself.

Figs. 5 and 6 further illustrate the representative behavior of the four planners from the perspectives of trajectory geometry and goal-approach dynamics. The three rows correspond to one preselected representative context from Categories A, B, and C, namely contexts 305, 795, and 986, respectively. Within each row, all methods are evaluated using the same initial and goal states, and the plotted results correspond to individual representative trials rather than category averages. The trajectory visualizations show that different planning paradigms can produce markedly different path-speed organizations under identical geometric conditions, while the goal-distance profiles further demonstrate that geometric approach to the goal and the formation of a stable completion state within the planning horizon are not equivalent. In particular, compared with MPD, ESAG reorganizes the goal-approach phase earlier while maintaining comparable geometric reachability, allowing the trajectory to enter and remain near the goal region earlier. This behavior is consistent with the overall improvements in T_{enter} and T_{sa} reported above.

Table II further reports the category-wise planning performance. In Category A, corresponding to regular midrange tasks, both MPD and ESAG achieve a navigation success rate

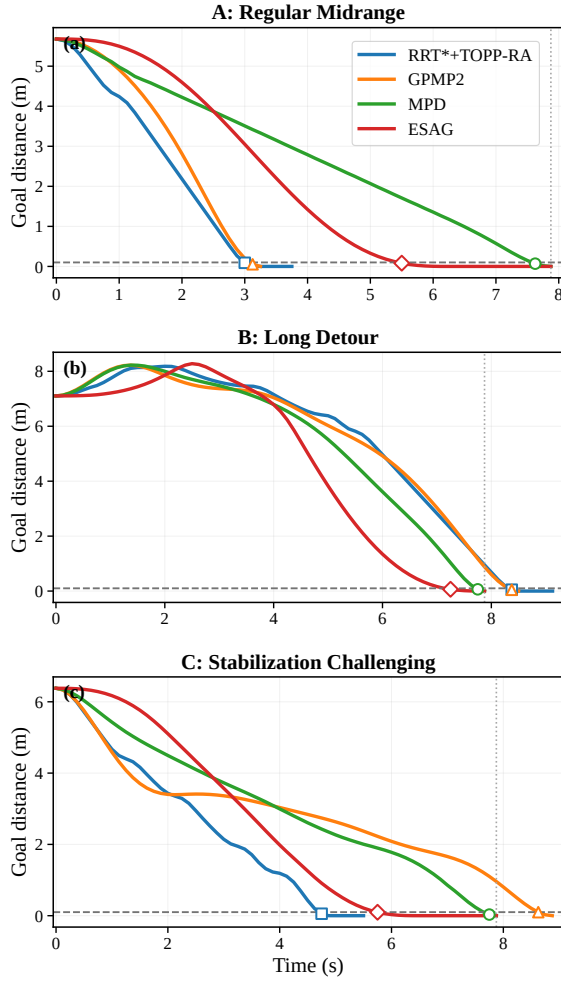


Fig. 6. Goal-distance profiles corresponding to the representative trials shown in Fig. 5. Markers indicate the first goal-entry time, while the dashed horizontal line denotes the goal-region threshold. The curves correspond to individual representative trials rather than averages over all contexts in each category.

of 100%, indicating that geometric reachability is not a dominant bottleneck in this category and enabling a direct comparison of stable-completion capability. The stable-completion rate of MPD is only 14%, whereas ESAG achieves 100%, corresponding to an absolute improvement of 86 percentage points. At the same time, the mean T_{sa} is advanced from 7.875 s to 6.135 s. By comparison, GPMP2-ATS achieves a navigation success rate of 99% but a stable-completion rate of only 3%, further demonstrating that successful entry into the goal region does not necessarily imply the formation of a persistently low-speed task-completion state. These results show that, under regular tasks with relatively sufficient temporal margin, ESAG can establish deceleration and stable terminal states earlier without degrading geometric navigation performance.

Category B, corresponding to long-detour tasks, imposes a substantially higher demand on the limited planning horizon. MPD achieves a navigation success rate of 82% but produces no trajectory satisfying the stable-completion criterion. ESAG achieves a navigation success rate of 80%, only 2 percentage points lower, while attaining a stable-completion rate of

TABLE II
SCENE-WISE PLANNING PERFORMANCE ON THE THREE CATEGORIES OF FIXED HELD-OUT SCENARIOS.

Scene	Method	Nav. (%)	Stable (%)	T_{enter} (s)	T_{sa} (s)
A	RRT*+TOPP-RA	91	91	3.286	3.523
	GPMP2-ATS	99	3	3.390	3.583
	MPD	100	14	7.628	7.875
	ESAG	100	100	5.566	6.135
B	RRT*+TOPP-RA	58	58	8.914	9.153
	GPMP2-ATS	56	1	8.556	13.250
	MPD	82	0	7.750	—
	ESAG	80	79	6.537	6.943
C	RRT*+TOPP-RA	69	69	5.149	5.388
	GPMP2-ATS	79	3	5.163	11.167
	MPD	95	13	7.696	7.875
	ESAG	95	95	6.079	6.665

79%. In addition, the mean T_{sa} values of RRT*+TOPP-RA and GPMP2-ATS reach 9.153 s and 13.250 s, respectively, reflecting the higher detour and temporal loads of this category. Notably, ESAG still achieves a high stable-completion rate without extending the fixed planning horizon. This indicates that its effect is not simply attributable to longer execution time, but rather to a more effective coordination of path detouring, goal approach, and terminal deceleration. Therefore, Category B serves as a stress test of the ability of ESAG to reorganize motion timing within a fixed horizon.

In Category C, where the main difficulty lies in stabilization near the goal, MPD and ESAG achieve the same navigation success rate of 95%. However, their stable-completion rates are 13% and 95%, respectively, yielding an absolute improvement of 82 percentage points for ESAG. The mean T_{sa} is also advanced from 7.875 s, located at the end of the fixed horizon, to 6.665 s. Because the two methods exhibit identical geometric navigation success in this category, the difference more directly indicates that the primary bottleneck is not finding a feasible path to the goal region, but suppressing residual motion and establishing a persistently stable terminal state during the final approach.

Across the three task categories, ESAG improves the stable-completion rate over MPD by 86, 79, and 82 percentage points in Categories A, B, and C, respectively, while the overall stable-completion rate increases from 9.00% to 91.33%. In contrast, the overall navigation success rate changes only from 92.33% to 91.67%, a difference of 0.66 percentage points. These results indicate that the primary contribution of ESAG is not to alter the basic geometric reachability of the diffusion planner, but to reorganize temporal allocation and terminal-state evolution within the fixed planning horizon, thereby converting a substantially larger fraction of successfully navigated trajectories into stably completed trajectories. The results on Categories A, B, and C respectively validate this effect under regular motion, long-detour and high-temporal-load conditions, and stabilization-dominated goal-approach tasks, which is consistent with the dominant difficulties characterized by the preceding task taxonomy.

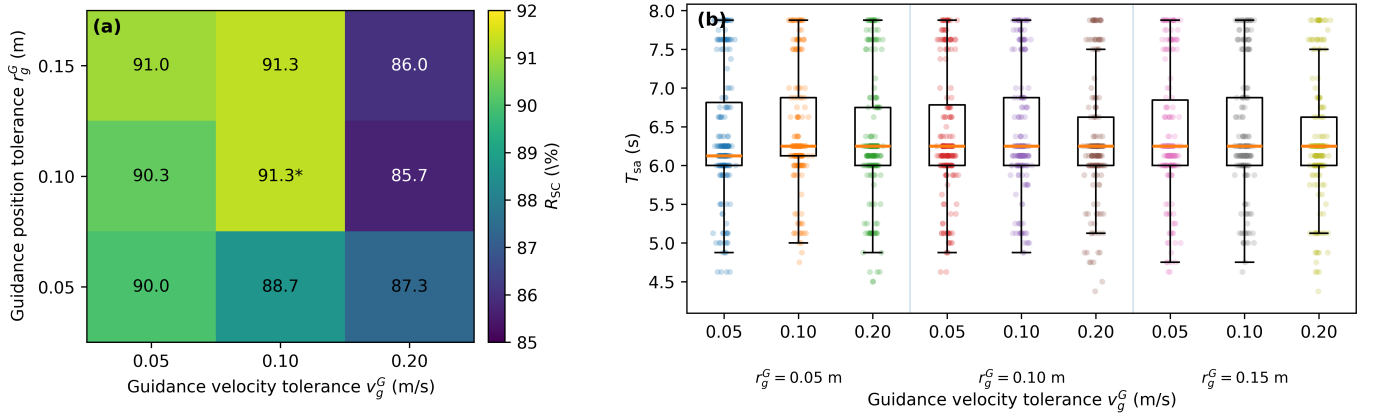


Fig. 7. Sensitivity of ESAG to the stable-goal tolerances used during guidance. (a) Stable-completion rate over the 3×3 guidance-tolerance grid. (b) Distribution of the stable-arrival time T_{sa} over successfully completed trials. The evaluation criterion is fixed at $(r_g^E, v_g^E) = (0.10 \text{ m}, 0.10 \text{ m/s})$ for all configurations, and the nominal guidance setting $(r_g^G, v_g^G) = (0.10 \text{ m}, 0.10 \text{ m/s})$ is marked by an asterisk in (a).

TABLE III
ROBUSTNESS OF ESAG TO THE GUIDANCE STABLE-GOAL TOLERANCES.

Metric	Nominal Setting	Minimum	Maximum	Range
Navigation success (%)	91.667	88.66	93.66	5.0
Stable completion (%)	91.33	85.663	92.997	7.333
Mean T_{sa} (s)	6.442	6.326	6.452	0.126

C. Sensitivity Analysis of the Stable-Goal Guidance Criterion

To investigate the dependence of ESAG on the stable-goal guidance parameters, the position tolerance r_g and velocity tolerance v_g used during guidance are varied separately while all other experimental settings are kept unchanged. The same evaluation protocol as that used in the main experiments is adopted. Fig. 7 and Table III report the planning performance of ESAG under different stable-goal guidance parameters. Overall, within the parameter ranges considered, the method does not exhibit a strong dependence on any single parameter combination. As summarized in Table III, the navigation success rate varies from 88.66% to 93.66%, while the stable-completion rate ranges from 85.663% to 92.997%. Under the nominal parameter setting, the two metrics are 91.667% and 91.33%, respectively. These results indicate that ESAG maintains relatively stable navigation feasibility and stable-completion capability over a reasonably broad range of stable-goal tolerances without scenario-specific parameter retuning.

A closer examination of Fig. 7(a) shows that the stable-completion rate is relatively more sensitive to the guidance velocity tolerance. When the velocity tolerance is set to 0.05 m/s or 0.10 m/s, the stable-completion rate remains generally high across different position tolerances. When the velocity tolerance is relaxed to 0.20 m/s, however, the stable-completion rate decreases for all tested position tolerances. In contrast, varying the position tolerance under the same velocity tolerance results in smaller changes, without a consistently monotonic improvement or degradation. This suggests that the position tolerance is not the dominant factor governing the stable-completion performance of ESAG, whereas an excessively relaxed velocity tolerance weakens the guidance

toward terminal states with low residual velocity during the reverse diffusion process. Consequently, some trajectories may enter the goal neighborhood without sufficiently establishing a persistently stable terminal motion state.

This observation is consistent with the design objective of ESAG. Stable completion requires not only that the trajectory reach the goal neighborhood, but also that the subsequent states exhibit sufficiently small residual motion and remain stable thereafter. Therefore, a moderately tighter velocity tolerance provides a clearer constraint on the terminal motion state during reverse diffusion, whereas an excessively relaxed velocity tolerance reduces the ability of the guidance criterion to distinguish trajectories with sufficiently stable temporal structures. Notably, this variation is primarily reflected in the probability of achieving stable completion and does not substantially alter the temporal location at which successful trajectories establish the stable state. As reported in Table III, the mean stable-arrival time varies only from 6.326 s to 6.452 s across different parameter combinations, corresponding to an overall variation of 0.126 s. Consistently, the stable-arrival-time distributions in Fig. 7(b) exhibit substantial overlap among the tested settings. These results indicate that variations in the stable-goal parameters primarily affect whether trajectories close to the decision boundary can form a sufficiently strict stable suffix, while the completion timing of trajectories that successfully establish a stable state remains largely unchanged.

Overall, ESAG exhibits good local robustness to the stable-goal guidance parameters. Within the tested range, the position tolerance has a limited influence on the overall performance, and the stable-arrival time remains nearly unchanged. The velocity tolerance mainly affects trajectories close to the stable-completion boundary, with an observable performance degradation when it is excessively relaxed. The nominal setting provides a balanced tradeoff among navigation success, stable-completion performance, and completion timing. This suggests that ESAG does not require fine-grained tuning of the stable-goal tolerances for individual scenarios to maintain stable performance, thereby supporting the use of a unified set of guidance parameters in the subsequent experiments.

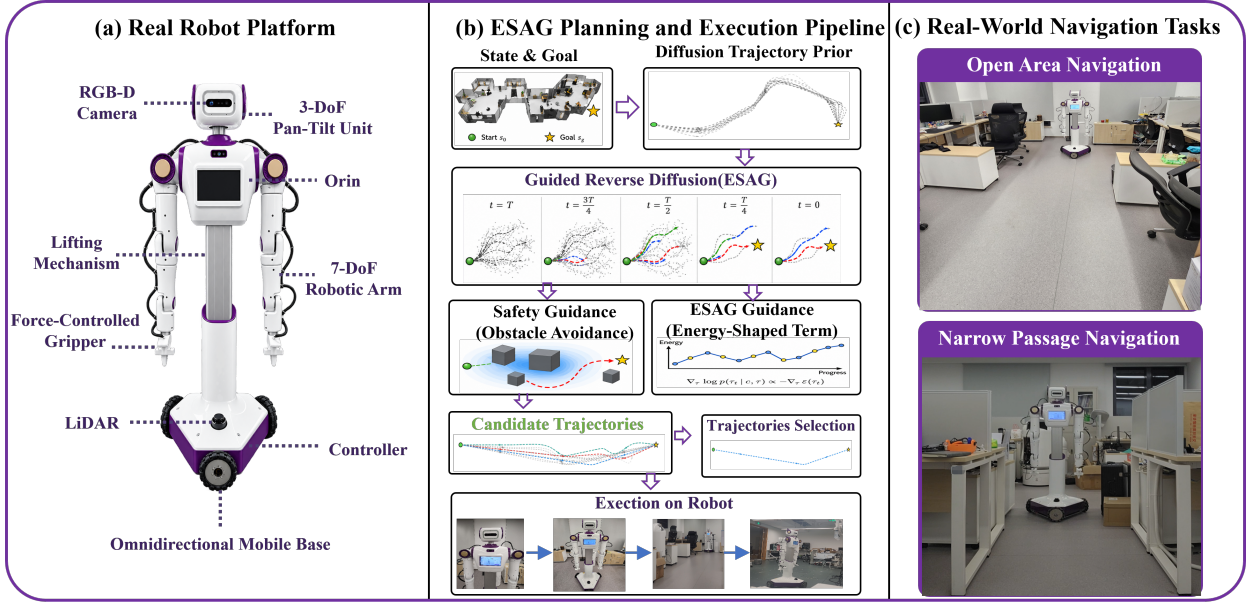


Fig. 8. Real-robot experimental setup for evaluating MPD and MPD+ESAG in representative indoor navigation scenarios.

TABLE IV
REAL-ROBOT PLATFORM AND PLANNING CONFIGURATION.

Parameter	Setting
<i>Robot platform</i>	
Mobile base	Omnidirectional
Footprint radius	0.35 m
Maximum velocity $v_{\max}$	1.5 m/s
Maximum acceleration $a_{\max}$	3.0 m/s ²
Base control frequency	20 Hz
<i>Planning setup</i>	
Planning horizon $T_{\max}$	8.0 s
Trajectory support points H	64
Temporal interval Δt	0.125 s
Goal radius r_g	0.10 m
Stable-speed threshold v_g	0.10 m/s

D. Real-Robot Experiments

To further validate the effectiveness of ESAG on a real robotic system, real-world navigation experiments are conducted in an indoor laboratory environment. The experiments use the mobile manipulation platform shown in Fig. 8, where only the omnidirectional mobile base is controlled for navigation, while the upper body and manipulators remain in a fixed safe configuration. The test scenarios include open-area navigation, narrow-passage navigation, and constrained goal-reaching tasks, allowing the navigation and stable-completion performance to be evaluated under different spatial constraints. The physical platform and main experimental parameters are summarized in Table IV. MPD and ESAG use the same robot model, planning horizon, motion constraints, and stable-goal criterion. During the experiments, the planned trajectories, executed trajectories, robot poses, and base velocities are recorded synchronously. The evaluation metrics include

navigation success rate, stable-completion rate, collision-free rate, T_{enter} , T_{sa} , executed path length, and trajectory-tracking RMSE.

Table V summarizes the overall real-robot results of MPD and ESAG. The navigation success rates of the two methods are 98.7% and 97.3%, respectively, while both maintain a 100% collision-free rate, indicating that ESAG largely preserves the original geometric navigation and collision-safety capabilities. In contrast, the stable-completion rate increases from 13.9% for MPD to 95.5% for ESAG, corresponding to an absolute improvement of 81.6 percentage points. This result indicates that, in the real system, successful entry into the goal region does not necessarily imply the formation of a stable task-completion state that can be continuously maintained.

The temporal metrics further reveal the difference between the two methods in the goal-approach and terminal-stabilization processes. The mean T_{enter} and T_{sa} of MPD are 7.575 s and 7.875 s, respectively, whereas ESAG reduces them to 5.427 s and 6.156 s. This corresponds to advances of 2.148 s in first goal entry and 1.719 s in stable completion. Meanwhile, the average executed path lengths of MPD and ESAG are 4.391 m and 3.986 m, respectively, while the trajectory-tracking RMSE values are 0.083 m and 0.077 m. These results indicate that the improvement in stable-completion performance is not achieved at the cost of additional detouring or degraded low-level tracking accuracy.

Fig. 9 further illustrates the planned trajectories and corresponding goal-distance profiles $d_g(t)$ of MPD and ESAG in several representative navigation scenarios. Both methods are able to generate feasible trajectories toward the goal region, whereas ESAG enters the goal neighborhood earlier in most cases, with the corresponding goal-distance profiles exhibiting earlier decrease and convergence. This observation is consistent with the improvement in T_{enter} reported in Table V. Together with the overall stable-completion rate and

TABLE V
OVERALL PERFORMANCE OF MPD AND ESAG ON THE SELECTED NAVIGATION CONTEXTS.

Method	Nav. (%) $\uparrow$	Stable (%) $\uparrow$	CF (%) $\uparrow$	T_{enter} (s) $\downarrow$	T_{sa} (s) $\downarrow$	Path (m) $\downarrow$	Track RMSE (m) $\downarrow$
MPD	98.7	13.9	100.0	7.575	7.875	4.391	0.083
ESAG	97.3	95.5	100.0	5.427	6.156	3.986	0.077

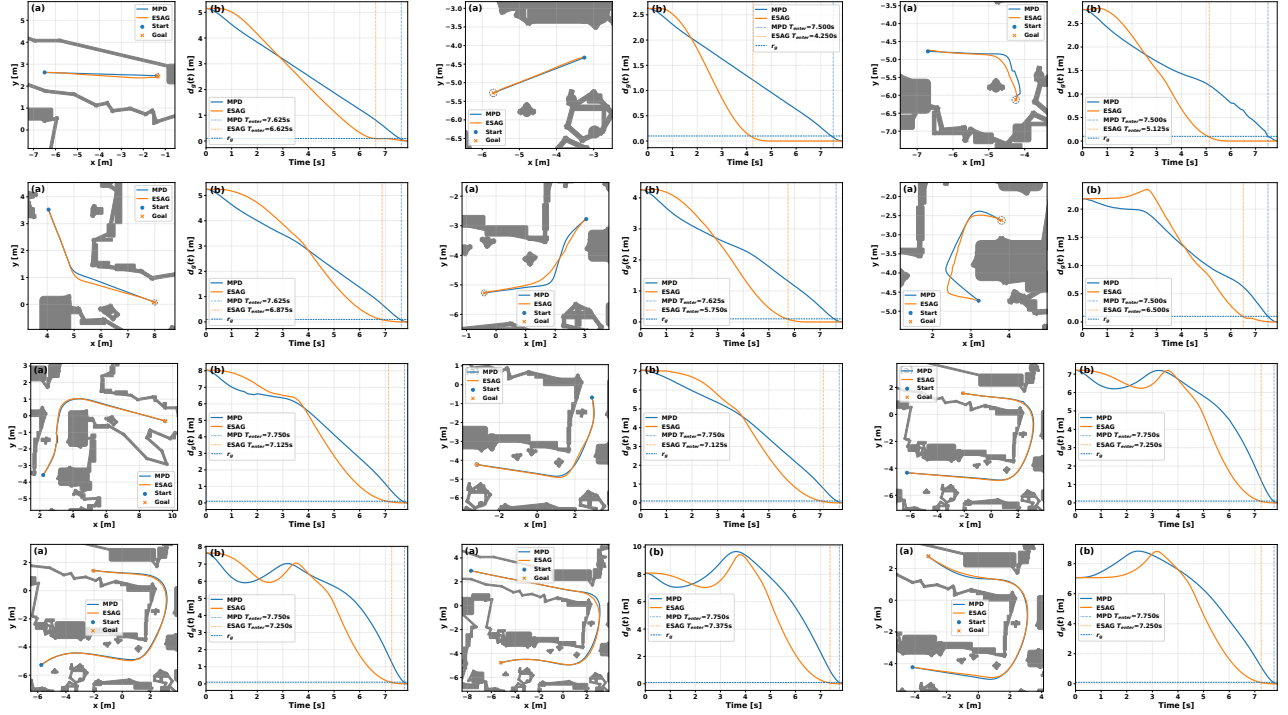


Fig. 9. Representative planned trajectories and goal-distance profiles of MPD and ESAG across different indoor navigation contexts. For each context, the left panel shows the planned trajectories, and the right panel shows the corresponding goal-distance profiles $d_g(t)$. The dotted horizontal line indicates the goal-region threshold r_g , and the vertical markers denote the first goal-entry time T_{enter} for MPD and ESAG.

T_{sa} results, these representative examples further indicate that ESAG not only improves the goal-approach process, but also enhances the formation of stable-completion states within the fixed planning horizon.

VI. CONCLUSION

This work investigates the explicit regulation of stable task-completion timing in fixed-horizon diffusion motion planning. We formulate task completion as the persistent satisfaction of goal-position and low-velocity conditions over a trajectory suffix and develop ESAG to optimize when such a stable completion state is established within the prescribed temporal support. By reformulating the discrete earliest stable-arrival time through suffix-wise incompleteness occupancy and constructing its differentiable relaxation, ESAG enables completion timing to participate directly in inference-time diffusion guidance without retraining the trajectory prior or modifying the planning horizon. Simulation and real-robot experiments demonstrate that ESAG substantially improves stable-completion performance and advances stable-arrival timing while largely preserving navigation success, collision safety, and motion feasibility. These results indicate that fixed-horizon generative planning can explicitly exploit the formation time of persistent

task completion as an additional temporal degree of freedom, which is particularly relevant to automation tasks in which navigation must reliably transition to subsequent operations. Future work will extend the formulation to dynamic environments and more complex robotic tasks, including mobile manipulation and multi-robot coordination.

REFERENCES

- [1] M. Görner, R. Haschke, H. Ritter, and J. Zhang, "Moveit! task constructor for task-level motion planning," in *2019 International Conference on Robotics and Automation (ICRA)*. IEEE, 2019, pp. 190–196.
- [2] D. Fox, W. Burgard, and S. Thrun, "The dynamic window approach to collision avoidance," *IEEE Robotics & Automation Magazine*, vol. 4, no. 1, pp. 23–33, 1997.
- [3] P. R. Wurman, R. D'Andrea, and M. Mountz, "Coordinating hundreds of cooperative, autonomous vehicles in warehouses," *AI Magazine*, vol. 29, no. 1, pp. 9–19, 2008.
- [4] J. Li, A. Tinka, S. Kiesel, J. W. Durham, T. K. S. Kumar, and S. Koenig, "Lifelong multi-agent path finding in large-scale warehouses," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 13, 2021, pp. 11 272–11 281.
- [5] B. Burger, P. M. Maffettone, V. V. Gusev, C. M. Aitchison, Y. Bai, X. Wang, X. Li, B. M. Alston, B. Li, R. Clowes, N. Rankin, B. Harris, R. S. Sprick, and A. I. Cooper, "A mobile robotic chemist," *Nature*, vol. 583, pp. 237–241, 2020.
- [6] M. Abolhasani and E. Kumacheva, "The rise of self-driving labs in chemical and materials sciences," *Nature Synthesis*, vol. 2, pp. 483–492, 2023.

- [7] M. Janner, Y. Du, J. B. Tenenbaum, and S. Levine, "Planning with diffusion for flexible behavior synthesis," in *Proceedings of the 39th International Conference on Machine Learning*, vol. 162. PMLR, 2022, pp. 9902–9915.
- [8] J. Carvalho, A. T. Le, M. Baierl, D. Koert, and J. Peters, "Motion planning diffusion: Learning and planning of robot motions with diffusion models," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2023, pp. 1916–1923.
- [9] J. Carvalho, A. T. Le, P. Kicki, D. Koert, and J. Peters, "Motion planning diffusion: Learning and adapting robot motion planning with diffusion models," *IEEE Transactions on Robotics*, vol. 41, pp. 4881–4901, 2025.
- [10] H. Huang, B. Sundaralingam, A. Mousavian, A. Murali, K. Goldberg, and D. Fox, "Diffusionseed: Seeding motion optimization with diffusion for rapid motion planning," in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 4392–4409.
- [11] M. Sharma, A. Fishman, V. Kumar, C. Paxton, and O. Kroemer, "Cascaded diffusion models for neural motion planning," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- [12] C. Pan, Z. Yi, G. Shi, and G. Qu, "Model-based diffusion for trajectory optimization," in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 57 914–57 943.
- [13] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," *The International Journal of Robotics Research*, vol. 44, no. 10-11, pp. 1684–1704, 2025.
- [14] A. Sridhar, D. Shah, C. Glossop, and S. Levine, "NoMaD: Goal masked diffusion policies for navigation and exploration," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 63–70.
- [15] B. Liao, S. Chen, H. Yin, B. Jiang, C. Wang, S. Yan, X. Zhang, X. Li, Y. Zhang, Q. Zhang, and X. Wang, "Diffusiondrive: Truncated diffusion model for end-to-end autonomous driving," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 12 037–12 047.
- [16] M. Yadegar, Z. Kashii, H. Ghaderi, N. Meskin, and M. Noorizadeh, "Autonomous target docking with obstacle avoidance and final velocity control for non-holonomic mobile robots," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 4039–4050, 2025.
- [17] Q.-C. Pham, "A general, fast, and robust implementation of the time-optimal path parameterization algorithm," *IEEE Transactions on Robotics*, vol. 30, no. 6, pp. 1533–1540, 2014.
- [18] H. Pham and Q.-C. Pham, "A new approach to time-optimal path parameterization based on reachability analysis," *IEEE Transactions on Robotics*, vol. 34, no. 3, pp. 645–659, 2018.
- [19] R. Liu, A. Hou, S. Li, and X. Yin, "VH-Diffuser: Variable horizon diffusion planner for time-aware goal-conditioned trajectory planning," *arXiv preprint arXiv:2509.11930*, 2025.
- [20] V. Sanjay, Y. Shaoul, and J. Li, "Diffusion for long-horizon multi-robot path planning in human-shared environments," *arXiv preprint arXiv:2607.09911*, 2026.
- [21] J. Guo, R. Zhang, Z. Li, Y. Gao, S. Peng, S. Lan, X. Hu, Z. Du, X. Zhang, and L. Li, "Efficient diffusion planning with temporal diffusion," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 26, 2026, pp. 21 450–21 458.
- [22] R. Liu, A. Hou, X. Yu, and X. Yin, "Zero-shot trajectory planning for signal temporal logic tasks," in *Advances in Neural Information Processing Systems*, vol. 38, 2025.
- [23] K. Saha, V. Mandadi, J. Reddy, A. Srikanth, A. Agarwal, B. Sen, A. Singh, and M. Krishna, "EDMP: Ensemble-of-costs-guided diffusion for motion planning," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 10 351–10 358.
- [24] Y. Luo, C. Sun, J. B. Tenenbaum, and Y. Du, "Potential based diffusion motion planning," *arXiv preprint arXiv:2407.06169*, 2024.
- [25] W. Xiao, T.-H. Wang, C. Gan, R. Hasani, M. Lechner, and D. Rus, "Safediffuser: Safe planning with diffusion probabilistic models," in *International Conference on Learning Representations*, 2025.
- [26] H. Ma, S. Bodmer, A. Carron, M. Zeilinger, and M. Muehlebach, "Constraint-aware diffusion guidance for robotics: Real-time obstacle avoidance for autonomous racing," in *Proceedings of the 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 305, 2025, pp. 1756–1776.
- [27] R. Romer, A. von Rohr, and A. P. Schoellig, "Diffusion predictive control with constraints," in *Proceedings of the 7th Annual Learning for Dynamics and Control Conference*, ser. Proceedings of Machine Learning Research, vol. 283, 2025, pp. 791–803.
- [28] W. Jung, U. A. Mishra, N. R. Arachchige, Y. Chen, D. Xu, and S. Kousik, "Joint model-based model-free diffusion for planning with constraints," in *Proceedings of the 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 305, 2025, pp. 4328–4350.
- [29] H. Jeon, C. Min, and J. Park, "Tree-guided diffusion planner," in *Advances in Neural Information Processing Systems*, vol. 38, 2025.
- [30] S. Yan, Z. Zhang, M. Han, Z. Wang, Q. Xie, Z. Li, Z. Li, H. Liu, X. Wang, and S.-C. Zhu, "M²Diffuser: Diffusion-based trajectory optimization for mobile manipulation in 3d scenes," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 1–17, 2025.
- [31] A. Srikanth, P. Mahajan, K. Saha, V. Mandadi, P. Paul, P. Wadhwani, B. Bhowmick, A. Singh, and M. Krishna, "GPD: Guided polynomial diffusion for motion planning," in *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)*. IEEE, 2025, pp. 2758–2765.
- [32] Y. Shaoul, I. Mishani, S. Vats, J. Li, and M. Likhachev, "Multi-robot motion planning with diffusion models," in *International Conference on Learning Representations*, 2025.
- [33] J. Liang, J. K. Christopher, S. Koenig, and F. Fioretto, "Simultaneous multi-robot motion planning with projected diffusion models," *arXiv preprint arXiv:2502.03607*, 2025.
- [34] Y. Zheng, R. Liang, K. Zheng, J. Zheng, L. Mao, J. Li, W. Gu, R. Ai, S. Li, X. Zhan, and J. Liu, "Diffusion-based planning for autonomous driving with flexible guidance," in *International Conference on Learning Representations*, 2025.
- [35] S. Samavi, A. Lem, F. Sato, S. Chen, Q. Gu, K. Yano, A. P. Schoellig, and F. Shkurti, "Sicnav-diffusion: Safe and interactive crowd navigation with diffusion trajectory predictions," *IEEE Robotics and Automation Letters*, vol. 10, no. 9, pp. 8738–8745, 2025.
- [36] Z. Xian, N. Gkanatsios, T. Gervet, T.-W. Ke, and K. Fragkiadaki, "Chaineddiffuser: Unifying trajectory diffusion and keypose prediction for robotic manipulation," in *Proceedings of the 7th Conference on Robot Learning*, vol. 229. PMLR, 2023, pp. 2323–2339.
- [37] X. Ma, S. Patidar, I. Haughton, and S. James, "Hierarchical diffusion policy for kinematics-aware multi-task robotic manipulation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- [38] C. Hao, A. Xiao, Z. Xue, and H. Soh, "CHD: Coupled hierarchical diffusion for long-horizon tasks," in *Proceedings of the 9th Conference on Robot Learning*, 2025, pp. 1420–1451.
- [39] D. Verscheure, B. Demeulenaere, J. Swevers, J. De Schutter, and M. Diehl, "Time-optimal path tracking for robots: A convex optimization approach," *IEEE Transactions on Automatic Control*, vol. 54, no. 10, pp. 2318–2327, 2009.
- [40] H. Huang, H. Liu, C. Xia, H. Mei, X. Gao, and B. Liang, "Sampling-based time-optimal path parameterization with jerk constraints for robotic manipulation," *Robotics and Autonomous Systems*, vol. 170, p. 104530, 2023.
- [41] S. Karaman and E. Frazzoli, "Sampling-based algorithms for optimal motion planning," *The International Journal of Robotics Research*, vol. 30, no. 7, pp. 846–894, 2011.
- [42] M. Mukadam, J. Dong, X. Yan, F. Dellaert, and B. Boots, "Continuous-time gaussian process motion planning via probabilistic inference," *The International Journal of Robotics Research*, vol. 37, no. 11, pp. 1319–1340, 2018.